

SIZE REDUCTION

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SIZE REDUCTION

- It is the process of reducing the particle size of a material to a finer state of subdivision (i.e. to smaller pieces, to coarse particles, or to powders).
- **Comminution is an alternative name for this operation**
- There are a number of methods which are used in size reduction
- E.g. cutting, chopping, crushing, or grinding depending on the type of equipment used and the material to be comminuted

SIZE REDUCTION...

■ Objectives of size reduction

- To produce smaller particles
- To reduce the bulk of materials
- To increase SA
- To enhance flowability of powders

■ Methods and Equipment of size reduction

A. Methods of size reduction

1. Cutting

- The material is cut by means of sharp blades

SIZE REDUCTION...

2. Compression

- In this method, the material is crushed by application of pressure

3. Impact

- Impact occurs when the material is more or less stationary and is hit by an object moving at high speed or when the moving particle strikes a stationary surface

4. Attrition

- Here the material is subjected to pressure as in compression, but the surface are moving relative to each others resulting in shear forces which break the particles

EQUIPMENT FOR SIZE REDUCTION

1. Cutter mill

Principle

In cutter mill, size reduction involves successive cutting with the help of sharp knives.

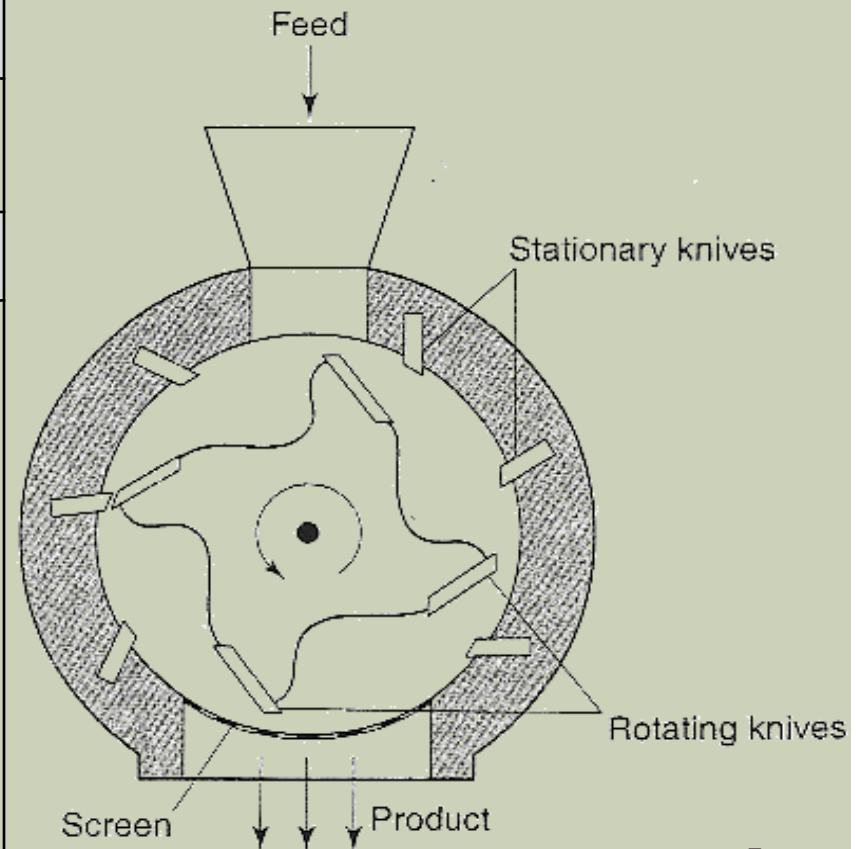
Working

Knives are attached to a rotor and act against stationary knives in the casing

Rotating knives are allowed to rotate at speed from 200 -900 rpm.

Feed loaded flows down by force of gravity.

During rotation, material comes very close between stationary & rotating knives & thereby cut materials into small pieces.



EQUIPMENT...

Use of Cutter mill

- ❖ To obtain coarse degree of size reduction of soft materials
- ❖ To reduce the size of tough & fibrous material.
E.g. roots, peels or woods prior to extraction

EQUIPMENT...

2. Roller mill

Principle

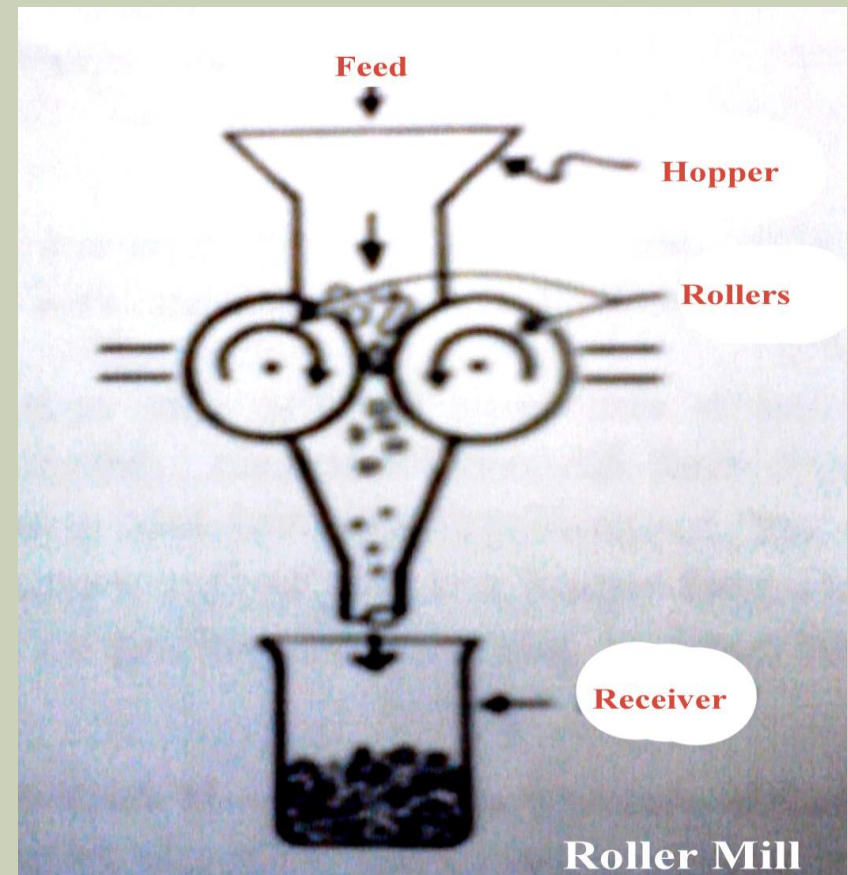
Material is crushed by application of stress through rotating heavy rollers.

Working

Rollers are allowed rotate in opposite direction to each other.
Rotating rollers crush materials with high pressure.

Use

It is used for crushing soft tissues of seeds before extraction of fixed oils.



EQUIPMENT...

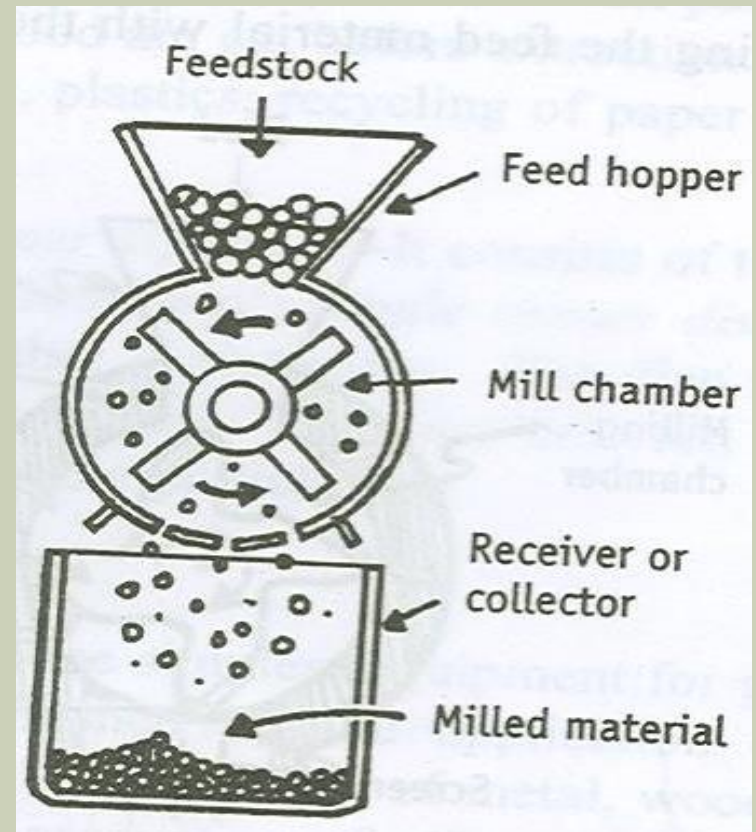
3. Hammer mill

Principle

Mill operates on principle of impact between rapidly moving hammers mounted on a rotor.

Working

Hammers are allowed to be in continuous motion (8000-15000 rpm). Rotating hammers beat materials to yield smaller particles.



EQUIPMENT...

Advantages of Hammer mill

- ❖ It is rapid in action and capable of grinding many d/t types of materials
- ❖ The product can be controlled by variation of rotor speed, hammer type and size and shape of mesh
- ❖ **Operation is continuous**
- ❖ Little contamination of the product with metal abraded from the mill

Disadvantages of Hammer mill

- ❖ The high speed of operation causes generation of heat that may affect thermolabile materials
- ❖ The mill may be chocked resulting decreased efficiency or even damage
- ❖ May be susceptible to damage by foreign objects in the feed

EQUIPMENT...

Use of Hammer mill

- ❖ It is capable of producing intermediate grades of powder from almost any substances except sticky materials
- ❖ It is used for non abrasive, brittle products.
- ❖ Applications include the powdering of barks, leaves, roots etc.

EQUIPMENT...

4. Ball mill

Principle

Material is crushed by impact between rapidly moving balls.

Working

The ball mill consists of a hollow cylinder mounted in such a way that it can rotate on its horizontal axis

Balls are made up of inert metals like steel will give impact action on material.

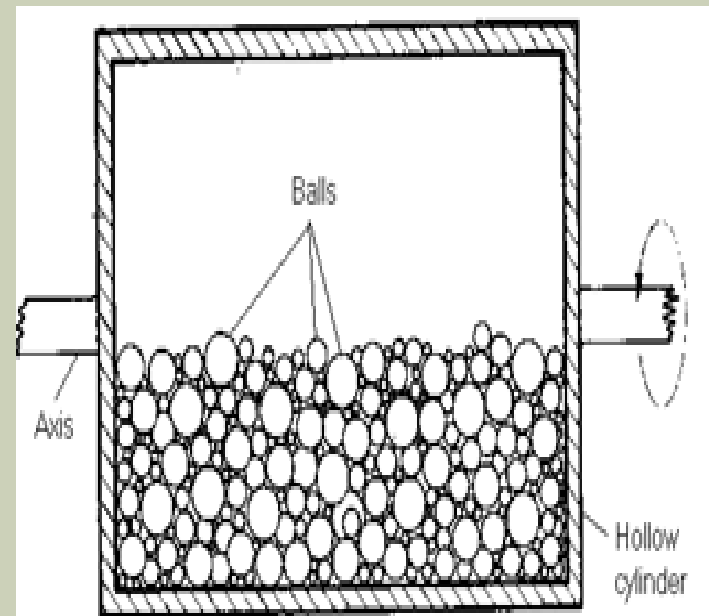


Fig. Ball mill

EQUIPMENT...

Cylinder:

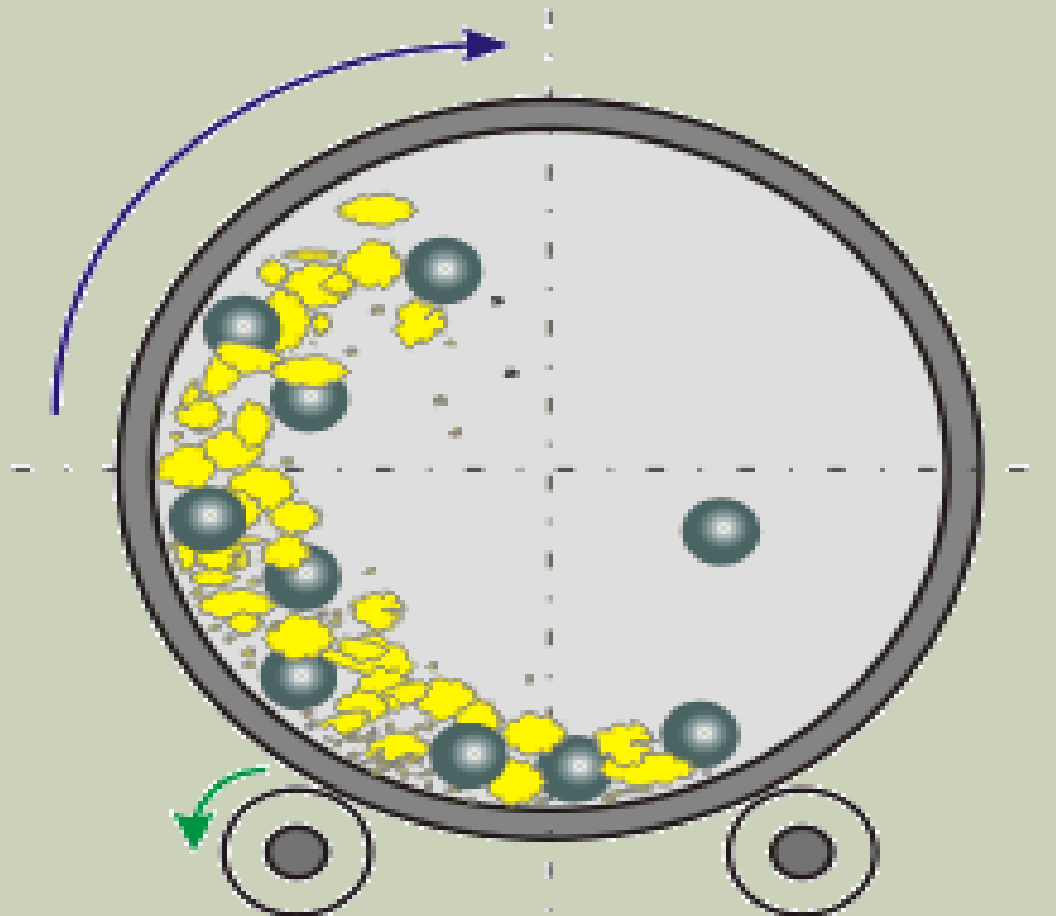
- Cylinder may be made up of metal, porcelain or rubber.
- Diameter of the cylinder ranges from **1-3m** in pharmaceutical practice.

Balls:

- Balls occupy about 30 to 50% of the volume of the cylinder.
- Diameter of the balls depends on the feed size and diameter of the cylinder.
- The diameter of balls ranges from **2-15cm**.
- Balls may be of metal, porcelain or pebbles

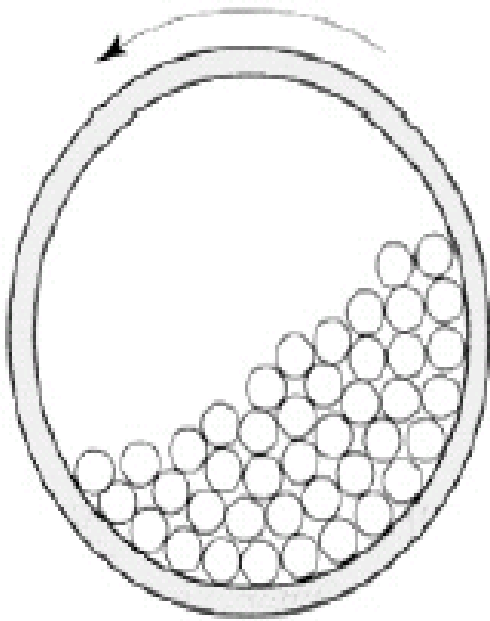
EQUIPMENT...

Here, speed of rotation of balls is very important.

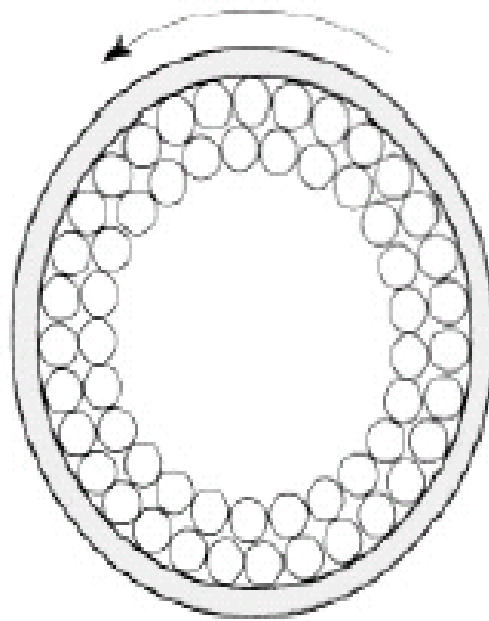


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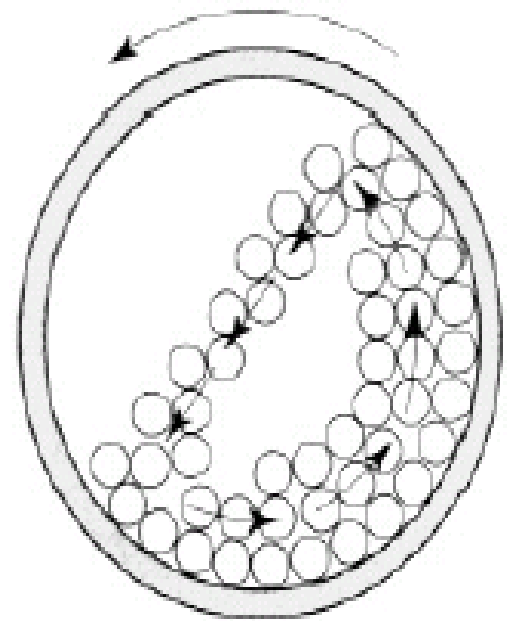
At low speed



At high speed



At moderate speed



EQUIPMENT...

Advantages of Ball mill

- ❖ Economical and simple to operate
- ❖ Can handle wide variety of materials of d/t character and degree of hardness
- ❖ It can produce very fine powders
- ❖ It is used for milling pigments etc

Disadvantages of Ball mill

- ❖ Wear and tear can occur
- ❖ Soft materials may stick on the sides of balls

EQUIPMENT...

5. Fluid energy mill

Principle

Feedstock is suspended within high velocity air stream.

Milling takes place because of high velocity collisions between suspended particles.

Working

After entering the powder from inlet, turbulent stream of air entering through the lower side & collide with suspended particles & break them.

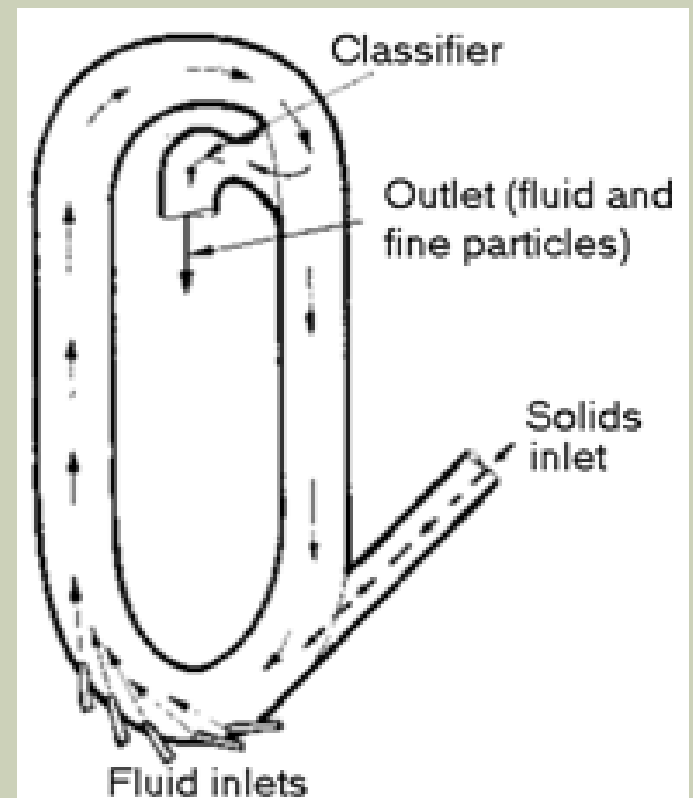


Fig. Fluid energy mill

EQUIPMENT...

- **Fluid energy mill** consists of a loop of pipe, which has a diameter of *2-20 cm*.
- The height of the loop may be up to **2m**.
- Several nozzles are fitted at the bottom of the pipe.
- A *classifier* is fitted in the system so that only finer size particles are collected as products and the larger size particles are again sent to the stream of air for further size reduction

EQUIPMENT...

Advantages of fluid energy mill

- ❖ Very small particle size of product is produced (up to 5 microns or less)
- ❖ Used for thermolabile substances
- ❖ Little or no abrasion of mill
- ❖ Used when especially fine powders are required
E.g. antibiotics & vitamins

EQUIPMENT...

Pharmaceutical applications of size reduction

- ❖ To increase SA of drug
- ❖ To increase therapeutic effectiveness of certain drugs
- ❖ To increase the stability of liquid preparations
- ❖ To increase the physical appearance of certain DFs
- ❖ To facilitate extraction of medicinal components from plants
- ❖ To facilitate drying
- ❖ For uniform distribution of medicaments in solid DFs